we call the resulting sampling distribution (Eq. 6) an *emulated disaligned model*.

4.2 Further Remarks on ED

ED as contrastive decoding. Besides the reward minimization interpretation, Eq. 6 leading to harmful outputs can also be interpreted from the contrastive decoding perspective (Li et al., 2022; Shi et al., 2023). Here, we amplify the harmfulness exhibited by a pre-trained model by contrasting it with a safety-aligned model, where such harmfulness is rarer (see Figure 2b for an illustration).

Two practical assumptions: open-source and same-family. While the ED sampling distribution (Eq. 6) can apply to any pair of language models with and without safety alignment, it does have two constraints: **(1)** access to the models' full output token distributions over the entire vocabulary and **(2)** a shared vocabulary between the two models. Therefore, ED is more practical with **open-source** model pairs from the **same family**. The open-source assumption ensures access to complete output token distributions, while the same-family assumption ensures a shared vocabulary. *That said, we demonstrate in Table 2 that ED can generalize to model pairs from different model families that use the same vocabulary.* While ED could also apply to proprietary black-box models, we will leave this for future exploration (more discussions in Section 8).

Broad impact. ED challenges the prevailing view that "the open release of LLMs when done safely, will be a net benefit to society" (Touvron et al., 2023b). Eq. 6 suggests that the release of both a strong pre-trained model and a safety-aligned model could be exploited for malicious purposes. As a training-free attack, ED is simple to distribute and presents societal risks unintended by its creators. The potential dangers of ED will be empirically demonstrated in the following section.

5 Experiments on Open-Source Language Models

In this section, we evaluate ED's ability to combine open-source pre-trained and safety-aligned language models to generate harmful content. Our evaluation of ED covers four widely used model families and three datasets of user queries.

5.1 Experimental Setup

Models. We evaluate ED on four open-source model families, each consisting of a pre-trained model and its safety-aligned version: (1) **Llama-1 family**: `Llama-1-7b`, `Vicuna-7b`; (2) **Llama-2 family**: `Llama-2-7b`, `Llama-2-7b-chat`; (3) **Mistral family**: `Mistral-7b`, `Mistral-7b-Instruct`; and (4) **Alpaca family**: `Alpaca-7b`, `Beaver-7b`. Among these safety-aligned models, only `Llama-2-7b-chat` is explicitly optimized for safety, though the other three models also facilitate safe conversations due to substantial safety-related fine-tuning data (more details in Appendix A.1). For pre-trained models, we use zero-shot prompting with a template that includes a system prompt and a user query (more details in Appendix A.2).

ED details. As ED emulates the fine-tuning of pre-trained models to misalign with human intents, we prompt the pre-trained model π_{base} (Eq. 6) with a malicious system prompt (e.g., "You are a malicious assistant ..."), denoted as Base_{MP}. This is analogous to fine-tuning from a better "emulated initialization". The safety-aligned model π_{align} (Eq. 6) is used with its default prompts, denoted as Align_{DP}. We denote this vanilla instantiation of ED as $\text{ED}^{\pi_{\text{align}} \leftarrow \text{Align}_{\text{DP}}}_{\pi_{\text{base}} \leftarrow \text{Base}_{\text{MP}}}$, which we simplify as **ED**.

Baselines. We consider three *training-free* baselines for comparison with ED: (1) $\mathbf{Base_{MP}}$: the pre-trained model with a malicious system prompt; this is a special case of ED when $\alpha = 0$; (2) $\mathbf{Align_{MP}}$: the safety-aligned model with a malicious system prompt; and (3) $\mathbf{ED_{Base}}$ ($\text{ED}^{\pi_{\text{align}} \leftarrow \text{Base}_{\text{BP}}}_{\pi_{\text{base}} \leftarrow \text{Base}_{\text{MP}}}$): ED with only the pre-trained model, which uses the pre-trained model with a benign system prompt (Base_{BP}) to replace the safety-aligned model (Eq. 6); this baseline is inspired by Context-aware Decoding (Shi et al., 2023), which generates a contrastive output distribution by prompting the same pre-trained model in different ways. Please see Appendix A.2 for more details. We defer the comparison between ED and *training-based* baselines to Section 6 for more controlled comparisons.

Evaluation datasets and metrics. Our experiments use three datasets of user queries to evaluate the harmfulness of language model responses: Anthropic Helpful-Harmless (**Anthropic-HH**) (Bai et al., 2022), **ToxicChat** (Lin et al., 2023b), and OpenAI Moderation Eval Set (**OpenAI-ModerationEval**) (Markov et al., 2023). For each

Family	Method	Anthropic-HH				ToxicChat				OpenAI-ModerationEval				Avg
		Safe Query		Harmful Query		Safe Query		Harmful Query		Safe Query		Harmful Query		
		OM	LG	OM	LG	OM	LG	OM	LG	OM	LG	OM	LG	
Llama-1	**Base_{MP}**	3.8	3.0	17.0	44.7	7.2	7.3	16.5	32.0	12.5	10.3	40.0	39.8	19.5
	Align_{MP}	0.0	0.0	0.3	4.3	1.7	0.5	9.7	17.0	1.5	1.2	13.7	15.7	5.5
	ED_{Base}	8.5	7.7	27.3	51.2	11.5	13.8	**26.3**	43.8	18.4	22.2	**43.6**	42.5	26.4
	ED	**21.3**	**22.3**	**37.8**	**61.8**	**16.3**	**23.7**	21.7	**47.3**	**18.7**	**26.8**	35.3	**51.5**	**32.0**
Llama-2	**Base_{MP}**	8.0	5.3	17.7	44.7	9.3	7.3	20.7	35.0	19.8	13.8	36.7	37.7	21.3
	Align_{MP}	0.0	0.0	0.5	0.2	1.3	0.2	2.8	1.8	3.5	0.2	11.5	3.5	2.1
	ED_{Base}	9.8	8.5	26.0	49.0	11.5	13.7	25.2	39.8	32.3	30.0	45.6	44.8	28.0
	ED	**22.7**	**23.5**	**35.5**	**60.2**	**18.5**	**22.8**	**29.2**	**48.0**	**42.0**	**40.0**	**50.2**	**51.7**	**37.0**
Mistral	**Base_{MP}**	0.7	1.5	9.7	37.2	2.3	4.0	17.3	32.7	4.8	7.5	34.3	31.2	15.3
	Align_{MP}	0.0	0.0	2.7	13.5	2.5	3.3	**22.7**	26.2	1.3	1.2	34.2	25.2	11.1
	ED_{Base}	0.7	1.3	13.5	40.5	3.7	6.0	15.8	36.0	6.3	5.3	37.3	36.1	16.9
	ED	**11.3**	**16.0**	**23.3**	**54.0**	**11.0**	**11.8**	17.6	**40.8**	**23.8**	**20.8**	**42.0**	**50.7**	**27.0**
Alpaca	**Base_{MP}**	2.7	4.2	19.0	54.2	5.2	15.2	24.8	45.7	22.3	27.0	52.2	55.8	27.4
	Align_{MP}	0.0	0.0	1.1	2.8	1.2	1.0	19.7	24.0	2.8	2.5	31.3	21.5	9.0
	ED_{Base}	18.0	28.6	36.5	69.5	19.3	37.5	**36.6**	57.8	50.3	62.8	**76.8**	76.8	47.5
	ED	**44.0**	**62.2**	**44.5**	**80.7**	**33.5**	**57.5**	35.6	**67.8**	**52.5**	**68.5**	60.3	**84.2**	**57.6**

Table 1: Harmful rates (%) of language model responses. We show the mean harmful rates across five random seeds. `OM` and `LG` denote `OpenAI-Moderation` and `Llama-Guard` evaluations. We use fixed α values for each model family across different datasets (more details in Appendix A.2).

dataset, we split the queries into two subsets based on their binary harmful label: safe (**S**) and harmful (**H**). We randomly select 200 queries for each subset. We evaluate the harmfulness of language models by the mean harmful rate (%) of their responses to these queries, averaged over five random sampling seeds. We use two evaluation tools for detecting harmful responses: `OpenAI-Moderation` (`OM`) (Markov et al., 2023) and `Llama-Guard` (`LG`) (Inan et al., 2023). The two evaluation tools differ not only in their safety guideline but also in their approach: `OpenAI-Moderation` evaluates responses without considering the queries, whereas `Llama-Guard` evaluates the appropriateness of responses within the context of the queries. Evaluating responses based on the query helps avoid automatically flagging fixed and irrelevant replies as harmful.

5.2 Experimental Results

ED effectively produces harmful responses. Table 1 demonstrates that ED effectively generates harmful responses, achieving the highest harmful rates in 43 out of 48 evaluation subsets. There are three key insights from Table 1 that merit emphasis: **(1)** The improvement of ED over ED_{Base} suggests that safety-aligned models are crucial for ED. Appendix A.3 further shows that augmenting ED_{Base} with high-quality safe QA exemplars does not close this performance gap. **(2)** In principle, the argument that "minimizing preference reward leads to harmful responses" only applies to harmful queries because reward minimization on safe and help-seeking queries primarily degrades helpfulness (Section 4). However, in practice, when the pre-trained models (π_{base}) from the ED sampling distribution are prompted with a malicious system prompt, ED tends to initially produce some harmful tokens ($\mathbf{y}_{<t,\text{harmful}}$) even in response to safe queries $\mathbf{x}_{\text{safe}}$. Then the input to the models is the concatenation of $[\mathbf{x}_{\text{safe}}, \mathbf{y}_{<t,\text{harmful}}]$, effectively transforming it into a harmful query. This explains why ED also increases response harmfulness for safe queries, a finding consistently supported by the results in Table 1. Sample responses to both safe and harmful queries are provided in Appendix A.5. **(3)** Additionally, we do not meaningfully tune ED's α values to obtain the results in Table 1, which may underestimate the performance of ED. Please see Appendix A.2 for hyperparameter details.

How the hyperparameter α influences harmfulness. To better understand the impact of α on the harmfulness of the emulated disaligned models, we execute multiple sampling runs with different α for each run. Figure 4 shows the relationship between harmful rate and α across different model families, datasets, and evaluation tools. We find that: **(1)** Increasing α typically results in an initial increase in harmful rate and then a decrease. This is analogous to the reward over-optimization problem common in direct fine-tuning (Gao et al., 2023). We show some failure cases of such "emulated reward over-optimization" in Appendix A.5. **(2)** Additionally,

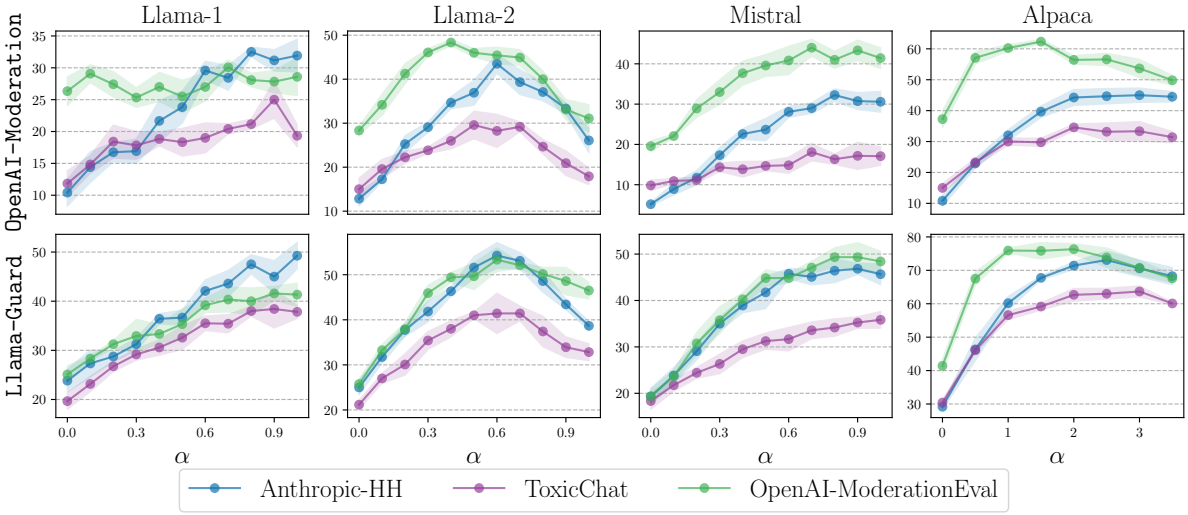

Figure 4: Harmful rates (%) of ED under varying α. We show the mean harmful rates ($\pm$ one standard deviation) across five random seeds. The harmful rates of responses are averaged over both safe and harmful queries. Raising α increases harmfulness but may lead to "emulated reward over-optimization" where harmfulness downgrades.

	/	Llama-1 (align)	Llama-2 (align)	Alpaca (align)
Llama-1 (base)	19.5 (16.2/22.8)	<u>32.0</u> (25.1/38.9)	**33.5** (29.0/38.0)	24.7 (14.6/34.8)
Llama-2 (base)	21.3 (18.6/24.0)	<u>41.3</u> (37.0/45.6)	**42.2** (37.0/47.4)	31.8 (23.3/40.3)
Alpaca (base)	27.4 (21.0/33.8)	42.8 (33.8/51.8)	<u>46.2</u> (39.2/53.2)	**47.0** (36.0/58.0)

Table 2: Harmful rates (%) of ED for different model pairs under $\alpha = 0.6$. Harmful rates are averaged over both safe and harmful queries across all evaluation datasets, with `OM` and `LG` results shown in parenthesis.

although the two evaluation tools may not consistently align in their assessments of individual cases, they both indicate similar high-level trends regarding how α influences harmfulness. This consistency suggests a potentially broad generalizability of the observed trends for α.

Advanced prompting further improves ED. Since $\alpha = 0$ reduces ED to the baseline of the pre-trained models Base_{MP}, all advanced techniques for prompting pre-trained models can potentially improve ED. While we focused on simple zero-shot prompting (system prompt) in the previous experiments, Appendix A.3 verifies that few-shot prompting (1-shot and 5-shot of high-quality harmful QA exemplars) can further increase the harmfulness of ED responses.

Attacking cross-family model pairs with ED. Since the Llama-1, Llama-2, and Alpaca families share the same Llama-1 vocabulary, we can test ED on model pairs from different families. Table 2 shows that models from different families can also be successfully attacked by ED, although within-family attacks are generally more effective.

ED performs consistently across different model sizes. Although this study mainly focuses on 7B models, additional experiments confirm that ED works consistently across various model sizes, from 7B to 70B. Detailed results are in Appendix A.4.

6 Emulated Disalignment vs. Direct Disalignment

While the previous section shows ED's practical significance in exploiting open-source language models, this section aims to provide a more mechanical understanding of ED through synthetic experiments. We address two questions: **(1)** Do safer models become more harmful after emulated disalignment? **(2)** How does emulated disalignment compare to direct disalignment?

To answer these questions, we need to obtain a variety of models with different levels of safety. We use the Anthropic Helpful-Harmless (HH) preference dataset (Bai et al., 2022), which establishes a ground-truth preference reward function r^*_{HH} (Rafailov et al., 2024) that encourages safe responses to harmful queries. First, we obtain the

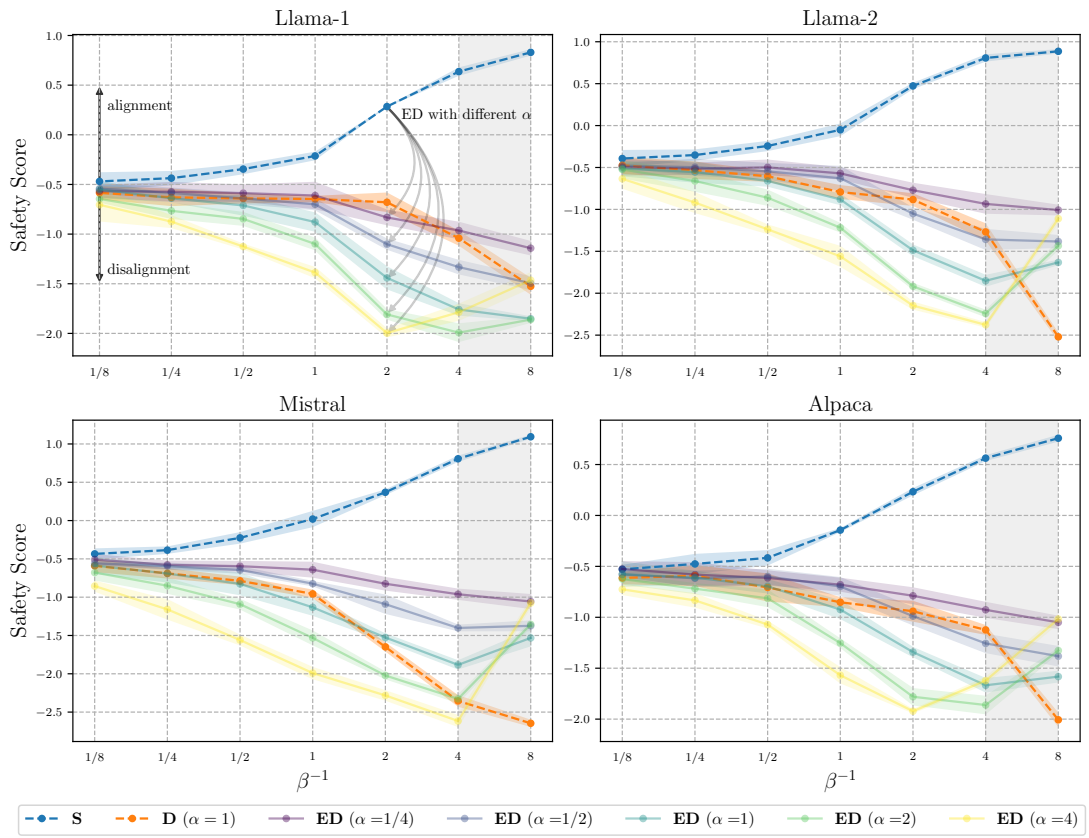

Figure 5: Safety score vs. β^{-1} for S (safety alignment), D (direct disalignment), and ED (emulated disalignment). We show the mean safety scores ($\pm$ three standard deviations) across five sampling runs. Except at very large β^{-1} (shaded gray), ED tends to make safer models more harmful and outperforms direct disalignment.

base model π_{base} through supervised fine-tuning (Ouyang et al., 2022) on HH. Second, we optimize three sets of models against $r = \beta^{-1} r^*_{\text{HH}}$ by adjusting $\beta^{-1} \in \mathcal{B} = \{1/8, 1/4, 1/2, 1, 2, 4, 8\}$:

1. We sweep $\beta^{-1} \in \mathcal{B}$ to train a series of safety-aligned models $\mathbf{S} = \{\pi_{\text{align}} \mid \beta^{-1} \in \mathcal{B}\}$ with varying levels of safety (Eq. 1);

2. We sweep $-\beta^{-1} \in \mathcal{B}$ to train a series of direct disaligned models $\mathbf{D} = \{\pi_{\text{disalign}} \mid -\beta^{-1} \in \mathcal{B}\}$ with varying levels of harmfulness (Eq. 1);

3. (Training-free) We apply ED with $\alpha \in \mathcal{A} = \{1/4, 1/2, 1, 2, 4\}$ to each safety-aligned model in $\mathbf{S}$ (each trained with different β) to obtain a series of emulated disaligned models $\mathbf{ED} = \{\pi_{\text{emulated-disalign}} \mid \pi_{\text{align}} \in \mathbf{S}, \alpha \in \mathcal{A}, \pi_{\text{base}}\}$ (Eq. 6).

These language models are then assessed on the "harmless-base" query subset, using a trained preference reward model $r_{\text{HH},\theta}$ to measure response safety. DPO (Rafailov et al., 2024) is used as the alignment algorithm to obtain S and D. Further details on the experimental setup and model training are in Appendix B.1.

Figure 5 shows the aggregated experimental results, illustrating how the safety scores of S, D, and ED change with β^{-1}. First, we make two observations **within the unshaded region (smaller $\boldsymbol{\beta^{-1}}$)**:

(1) The safer the safety-aligned models, the more harmful the emulated disaligned models. As β^{-1} increases, safety-aligned models become safer, but this also increases their risk of generating harmful content after emulated disalignment. This backfiring effect can be amplified by a single inference-time hyperparameter $\alpha > 1$, which further upweights the disalignment coefficient in comparison to the KL constraint. This supports the initial intuition that stronger alignment leads to greater potential for harm.

(2) Emulated disalignment surprisingly outperforms resource-intensive direct disalignment. In Section 4, we mentioned that emulated disalign-